



Image Enlargement by Lagrange Interpolation

Yung-Gi Wu

To cite this article: Yung-Gi Wu (2010) Image Enlargement by Lagrange Interpolation, International Journal of Computers and Applications, 32:4, 420-427

To link to this article: <https://doi.org/10.2316/Journal.202.2010.4.202-2470>



Published online: 11 Jul 2015.



Submit your article to this journal [↗](#)



Article views: 10



View related articles [↗](#)

IMAGE ENLARGEMENT BY LAGRANGE INTERPOLATION

Yung-Gi Wu*

Abstract

Nowadays, consumer electronics products are popular. Digital cameras have become the necessities of life. Due to the constraints of memory size or imaging resolution, image enlargement becomes very important in the field of image processing which can raise the effectiveness of displaying or printing image at the resolution higher than imaging. When the disk space for imaging device is limited, the enlargement technique can be used to raise the resolution of small pictures. The main objective of this paper is to scale the images thereby maintaining the quality without visible aliasing or distortion artifact. Interpolation is required in a variety of image processing applications. We use the Lagrange interpolation formula to constitute an interpolation polynomial to describe the tendency of image data and to estimate the unknown data to achieve the image enlargement. We compare the proposed method with existing image processing software, and the experimental results demonstrate that our method can efficiently enlarge the image with less complexity and better visual quality.

Key Words

Lagrange interpolation, estimate, image enlargement

1. Introduction

The consumer electronic products are more and more popular nowadays. Digital image/video processing is a progressive research area for multimedia applications due to the efficiency of digital representation of signals. Digital cameras and digital videos have become the essentials in daily life so that image processing is important because it is like a bridge between the natural image and users. When we catch an image from the camera, the image processing software can be used for enhancement, filtering, compression, scaling, and so on to meet the requirements of users. Image/video scaling is desirable in systems with different display resolutions. Many papers have been devoted to better algorithms to enlarge images. However, most of the tools fail to achieve good results when both visual quality and computation are considered.

Because the resolution or the disk storage provided in the digital camera is not high or huge enough, people always use the tool in the software to enlarge the image to see the detailed content of the image. Therefore, the enlargement of image is important. Nonetheless, image enlargement usually causes the change of image resolution. Digital image enlargement is generally obtained by scaling the image in a uniform method. This procedure is widely used in diverse areas of computer graphics and image processing. In general, we must develop some methods to insert the pixels to the new spaces among the original pixels when the image is enlarged. As Fig. 1 indicates, I is the original image and I' is the enlarged image. The dark blocks denote the original data and white blocks denote the new data to be inserted. Several methods for carrying out image scaling have been proposed and adopted in image processing packages, including nearest neighbour, bilinear interpolation, and bicubic interpolation. We will describe and compare those methods in the later sections.

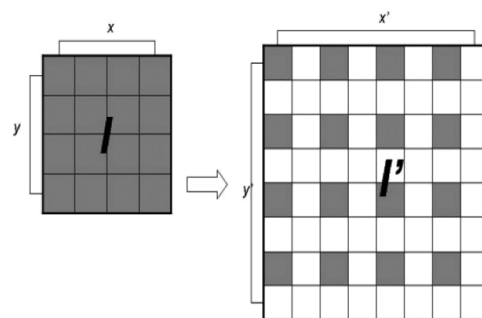


Figure 1. An example of image enlargement.

Nearest Neighbour uses the near image data to insert the new space. The advantage of this method is the low time complexity. However, the quality of the yielded image is poor. Another method called bilinear interpolation uses bilinear interpolation to estimate unknown data to insert the new space. It can easily achieve the function of image enlargement, but it suffers from the blocking effect. The other method called bicubic interpolation uses nearest 16 pixels to estimate and insert the new space by calculating the cubic interpolation kernel function twice to yield the data to pad. The advantage of this method is that it has a better quality with respect to visual judgment than other methods, but it needs more time complexity.

* Department of Computer Science & Information Engineering,
Chang Jung Christian University, Tainan, Taiwan; e-mail:
wyg@mail.cjcu.edu.tw

Recommended by Dr. L. C. Monticone
(10.2316/Journal.202.2010.4.202-2470)

The Lagrange interpolation formula was first published by Waring in 1779, rediscovered by Euler in 1783, and published by Lagrange in 1795 [1–3]. When interpolating polynomials are constructed, there is a tradeoff between having a better fit and having a smooth well-behaved fitting function. The more the data points are used in the interpolation, the higher the degree of the resulting polynomial, and therefore the greater oscillation it will exhibit between the data points. As a result, a high-degree interpolation is a poor predictor of the function between points, although the accuracy at the data points will be “perfect”. In the proposed method, we select the nearby four data to construct the interpolation function.

The Lagrange interpolating polynomial $P(x)$ of degree $n - 1$ that passes through the n points is given by

$$P(x) = \sum_{i=1}^n P_i(x), \quad (1)$$

where

$$P_i(x) = y_i \prod_{k=1, k \neq i}^n \frac{x - x_k}{x_i - x_k}. \quad (2)$$

Written explicitly,

$$\begin{aligned} P(x) = & \frac{(x - x_2)(x - x_3) \dots (x - x_n)}{(x_1 - x_2)(x_1 - x_3) \dots (x_1 - x_n)} y_1 \\ & + \frac{(x - x_1)(x - x_3) \dots (x - x_n)}{(x_2 - x_1)(x_2 - x_3) \dots (x_2 - x_n)} y_2 \\ & + \dots + \frac{(x - x_1)(x - x_2) \dots (x - x_{n-1})}{(x_n - x_1)(x_n - x_2) \dots (x_n - x_{n-1})} y_n \end{aligned} \quad (3)$$

The major aim of this paper is using this Lagrange interpolating polynomial to enlarge the image. In Section 2, we depict the other methods and Lagrange interpolating polynomial to enlarge image. In Section 3, we show the results of image enlargement by using the above methods. The conclusion of this paper is in Section 4.

2. Image Enlargement Techniques

In many cases, we have to enlarge a small image. When the desired size of the image is larger than the sampling resolution, we must insert the values to the new space by using the original image data in the proposed method. Many functions provided in the image processing tools can be used to enlarge image, but the quality of enlarged image is not good enough or the computation complexity is too expensive. Some popular methods, such as nearest neighbour, bilinear, and bicubic, are used in image processing software and also realized by hardware circuits to equip in electronic consumer products. The four published methods aimed to enlarge images are [4–7]: [4] is a fractal-based enlargement; [5] enlarges the image by estimating the high-frequency components of an interpolation-based

image; [6] uses Laplacian pyramid to predict the unknown high-frequency component to fulfill enlargement; and [7] employs grey interpolation to enlarge image. Among all the methods mentioned above, [5] and [6] have the heavier computation burden due to the frequency domain processing compared to the spatial domain processing. Fractal-based enlargement like [4] is a time-consuming scheme for the processing to find out the fractal codes (S_i , O_i , etc.). If the fractal codes are given, the time to fulfill enlargement will be very short. Interpolation-based method is fast in general, and the time is determined by the interpolation polynomial. All the four interpolation polynomials of bilinear, bicubic, Grey and Lagrange are not that complicated compared to the schemes of [4–6]. Nearest neighbour scheme has the least computation burden because it just needs to find out the shortest data point without complicated computation of polynomial. The interpolation method simulates a continuous curve function $P(x)$ by passing the discrete sample points that are composed by the original image data. Then, it uses this continuous curve function, or the interpolation function, to generate the values to be inserted. A paper which applies the Lagrange interpolation to fulfill the indexes concealment in image vector quantization is [8]. The following sections describe three published methods including nearest neighbour, bilinear interpolation, and bicubic interpolation. Our proposed Lagrange interpolation method will be given in Section 2.4.

2.1 Nearest Neighbour

The nearest neighbour interpolation is the simplest method in image scaling. Figure 2 shows the sample of nearest neighbour interpolation and the enlarged image. A , B , C , D are the four original pixels. The white blocks are the space to be predicted. Here, we pick point M to be our explanation for insertion in enlargement. To find the pixel value of M , we calculate the distances of point A to point M , point B to point M , point C to point M , and point D to point M , respectively. The point with shortest distance determines the pixel value of M . Refer to Fig. 2; the shortest distance to point M is from point A and the value of M is inserted by the value of A in this case.

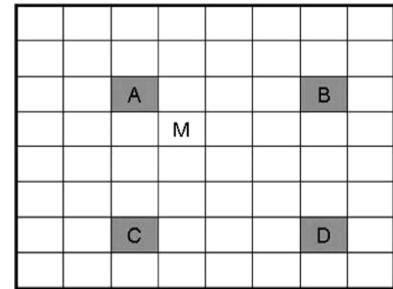


Figure 2. The example of nearest neighbour interpolation.

2.2 Bilinear Interpolation

The method of bilinear interpolation uses the nearest 2×2 pixels to calculate the pixel value of new space. The

example of bilinear interpolation is shown in Fig. 3. A , B , C , D are the four original pixels. α , β , γ , and δ denote the distances between points. The steps of calculating the pixel value of M by bilinear interpolation are given as follows:

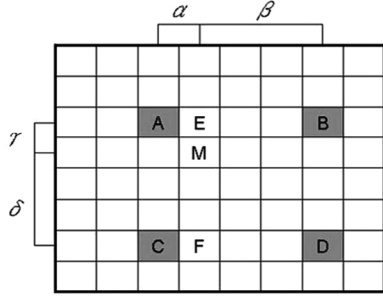


Figure 3. The example of bilinear interpolation.

Step 1: Calculate the pixel value of point E

$$E = \frac{\beta}{\alpha + \beta}A + \frac{\alpha}{\alpha + \beta}B \quad (4)$$

Step 2: Calculate the pixel value of point F

$$F = \frac{\beta}{\alpha + \beta}C + \frac{\alpha}{\alpha + \beta}D \quad (5)$$

Step 3: Calculate the pixel value of point M

$$M = \frac{\delta}{\gamma + \delta}E + \frac{\gamma}{\gamma + \delta}F \quad (6)$$

2.3 Bicubic Interpolation

The bicubic interpolation uses the nearest 4×4 pixels to calculate new pixel value in the scaled image. The example of bicubic interpolation is given in Fig. 4. α , β , p , and q denote the distances between data points.

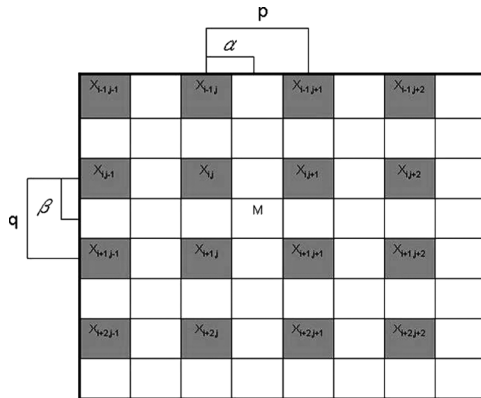


Figure 4. The example of bicubic interpolation.

The bicubic interpolation kernel function $h_c(x)$ is given as follows:

$$h_c(x) = \begin{cases} 1 - 2|x|^2 + |x|^3, & 0 \leq |x| < 1 \\ 4 - 8|x| + 5|x|^2 - |x|^3, & 1 \leq |x| < 2 \\ 0, & 2 \leq |x| \end{cases} \quad (7)$$

If M is the new value to be inserted, we have to use the nearest 16 pixels to convolute with $h_c(x)$ to get the value of M as follows:

$$M = \sum_{m=-1}^2 \sum_{n=-1}^2 X_{i+m,j+n} h_c\left(n - \frac{\beta}{q}\right) h_c\left(-\left(m - \frac{\alpha}{p}\right)\right) \quad (8)$$

where $X_{i+m,j+n}$ denotes the pixel value of original image. It is obvious that the proposed method has the highest complexity.

Four methods to achieve image enlargement are declared in [4–7]. In [4], a fractal-based image enlargement technique is proposed. This technique can preserve the details in edge regions while maintaining the smoothness in flat regions. This method needs to find the enhancement layer for decoding to reduce the difference between the range block and its associated transformed domain block. However, the computational burden is heavy because it needs extra computation to deal with the artifact of the blocky effect.

A method to finely enlarge an image by estimating the lost high-frequency components of an interpolation-based enlarged image is proposed in [5]. The estimation is accomplished by a self-decomposed codebook and inquiries based on the Mahalanobis distance. In [6], the Laplacian pyramid is used to hierarchically express the image. Based on Laplacian pyramid representation, the prediction of unknown higher-frequency components is equivalent to the prediction of an unknown high-resolution Laplacian image. In this paper, they change the bandwidth of the Gaussian filter to improve the performance of the enlargement method based on Laplacian pyramid representation. They show that the emphasis degree is changed as the bandwidth of the Gaussian filter is altered. An image enlargement technique based on grey polynomial interpolation is presented in [7]. It consists of three steps: (i) preprocess the data by the first-order accumulated generating operation (1-AGO); (ii) model a quadratic polynomial using a-AGO preprocessed data; and (iii) acquire interpolated pixels from the quadratic polynomial with the first-order inverse accumulated generating operation (1-IAGO). The results in [7] are better than bicubic method; however, the computation complexity includes the generation of accumulated generation, solve the quadratic polynomial, and inverse accumulated generation. Thus, the operation burden is heavy.

3. The Proposed Method

In this section, the proposed method is given. We constitute the Lagrange interpolation and linear interpolation to enlarge image. Figure 5 shows the example of our method. Here, the corresponding coordinates of those reference points are listed in Table 1. α , β , γ , and δ denote

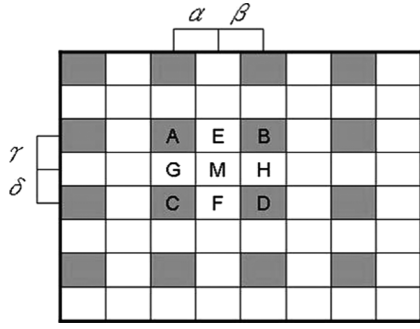


Figure 5. The example of our method.

Table 1
Corresponding Coordinates for Each Data

y_i	L_1	L_2	M	R_2	R_1
x_i	-2	-1	0	1	2

the distances between points. M denotes the unknown data in position $[m, n]$, and L_1, L_2, R_2, R_1 are the values of the four original reference points.

Constituting a polynomial $P(x)$ using the original four points is given as follows:

$$P(x) = \frac{(x+1)(x-1)(x-2)}{-12}L_1 + \frac{(x+2)(x-1)(x-2)}{6}L_2 + \frac{(x+2)(x+1)(x-2)}{-6}R_2 + \frac{(x+2)(x+1)(x-1)}{12}R_1 \quad (9)$$

Here, we generate two polynomials of $P_h(x)$ and $P_v(y)$ which mean the horizontal and vertical directions, respectively, to estimate the value of M in the position of $[m, n]$. Table 2 and Fig. 6 show the positions of reference points in each direction. By using (9) and Table 2, we can get two polynomials of $P_h(x)$ and $P_v(y)$ as given in (10a) and (10b). The final estimated value is determined by (11). W is the 3×3 window as shown in Fig. 6. $W[i, j]$ denotes the value in position $[i, j]$. $m-1 \leq i \leq m+1$ and $n-1 \leq j \leq n+1$

$$P_h(x) = \frac{(x+1)(x-1)(x-2)}{-12}W[m-1, n-1] + \frac{(x+2)(x-1)(x-2)}{6}W[m, n-1]$$

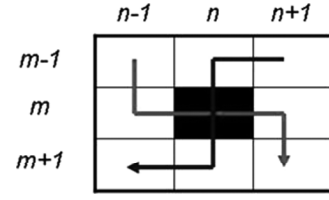


Figure 6. Two ways for interpolation.

$$+ \frac{(x+2)(x+1)(x-2)}{-6}W[m, n+1] + \frac{(x+2)(x+1)(x-1)}{12}W[m+1, n+1] \quad (10a)$$

$$P_v(y) = \frac{(y+1)(y-1)(y-2)}{-12}W[m-1, n+1] + \frac{(y+2)(y-1)(y-2)}{6}W[m-1, n] + \frac{(y+2)(y+1)(y-2)}{-6}W[m+1, n] + \frac{(y+2)(y+1)(y-1)}{12}W[m+1, n-1] \quad (10b)$$

The final value of predicted M is given by:

$$M = \frac{P_h(0) + P_v(0)}{2} \quad (11)$$

Our proposed method combines linear interpolation and Lagrange interpolation. Assume the position of M is $[m, n]$. The steps of our method are given as follows:

Step 1: Calculate the pixel value of point E

$$E = \frac{\beta}{\alpha + \beta}A + \frac{\alpha}{\alpha + \beta}B \quad (12)$$

Step 2: Calculate the pixel value of point F

$$F = \frac{\beta}{\alpha + \beta}C + \frac{\alpha}{\alpha + \beta}D \quad (13)$$

Step 3: Calculate the pixel value of point G

$$G = \frac{\delta}{\gamma + \delta}A + \frac{\gamma}{\gamma + \delta}C \quad (14)$$

Table 2
The Positions of Reference Point

y_i	L_1	L_2	M	R_2	R_1
Horizontal	$[m-1, n-1]$	$[m, n-1]$	$[m, n]$	$[m, n+1]$	$[m+1, n+1]$
Vertical	$[m-1, n+1]$	$[m-1, n]$	$[m, n]$	$[m+1, n]$	$[m+1, n-1]$

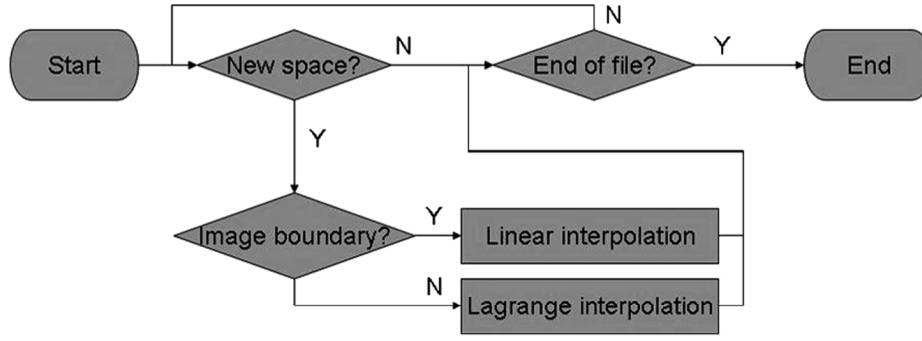


Figure 7. System diagram of image enlargement.

Step 4: Calculate the pixel value of point H

$$H = \frac{\delta}{\gamma + \delta}B + \frac{\gamma}{\gamma + \delta}D \quad (15)$$

Step 5: Constitute the Lagrange interpolation function

$$\begin{aligned} P_h(x) = & \frac{(x+1)(x-1)(x-2)}{-12}A \\ & + \frac{(x+2)(x-1)(x-2)}{6}G \\ & + \frac{(x+2)(x+1)(x-2)}{-6}H \\ & + \frac{(x+2)(x+1)(x-1)}{12}D \end{aligned} \quad (16a)$$

$$\begin{aligned} P_v(y) = & \frac{(y+1)(y-1)(y-2)}{-12}B \\ & + \frac{(y+2)(y-1)(y-2)}{6}E \\ & + \frac{(y+2)(y+1)(y-2)}{-6}F \\ & + \frac{(y+2)(y+1)(y-1)}{12}C \end{aligned} \quad (16b)$$

Step 6: Calculate the pixel value of point M

$$\begin{aligned} M = \frac{P_h(0) + P_v(0)}{2} = & \left(\left(-\frac{1}{6}A + \frac{2}{3}G + \frac{2}{3}H - \frac{1}{6}D \right) \right. \\ & \left. + \left(-\frac{1}{6}B + \frac{2}{3}E + \frac{2}{3}F - \frac{1}{6}C \right) \right) / 2 \end{aligned} \quad (17)$$

An exception does not use the abovementioned steps. If the position of M is located in the image boundary, we use linear interpolation to estimate it, because the image boundary could not get enough reference points to constitute the Lagrange interpolation polynomial as we described in previous sections. In this case, we use linear interpolation to deal with them. Figure 7 shows the configuration of the whole system.

Let I denote the original image, I' expresses the enlarged image, and $f(\cdot)$ denotes the zooming function, which contains (12) to (17). Zooming factor is 2. The following

expression is used to denote the relationship among I , $f(\cdot)$ and I' .

$$I' = f(I) \quad (18)$$

We can feed the I' to $f(\cdot)$ in the second round to get the bigger image size with the zooming factor of 4 compared to the original image I . The zoom factor in this paper is 2^n where $n \in \mathbb{Z}$.

Note that, compared to the three popular methods provided in image processing tools, the computation burden of the proposed method is less than bicubic method but greater than both bilinear and nearest neighbour methods. However, the quality of the proposed method outperforms the other methods.

4. Experimental Results

We select the five grey images, Boat, F16, Fruit, Lena, and Tiffany to test. All the original resolution of the test images is 512×512 . Our programming uses Microsoft visual studio.net C# and AMD K7 2500+ PC. To evaluate the performance, we compare the proposed method to the other methods by PSNR value and visual judgement. In the first experimental test, we re-sample image size to 256×256 from the original image and enlarge it to 512×512 by different methods including bicubic, bilinear, nearest neighbour [4-7] and Lagrange interpolation to evaluate the performance. The bicubic, bilinear, and nearest neighbour used for comparison are from the PhotoImpact 7.0. Table 3 shows the PSNR values between the reconstructed image and original image whose size is 512×512 . Some of the experimental results yielded by [4-7] are better than the proposed method. However, when we consider the computation burden, the proposed method has the lowest cost while maintaining the comparable enlarged image quality compared to the results of [4-7]. Table 4 shows the average time to enlarge five images from 256×256 to 512×512 in different schemes. Note that the time to yield fractal codes in [4] is not included.

Figure 8 is the original Lena image. Figures 9-12 are the reconstructed images by bicubic, bilinear, nearest neighbour, and the proposed method, respectively. The PSNR values are also provided. To evaluate the performance of the four methods, we also show the different

Table 3
The Quality of Reconstructed Images Expressed in dB by Different Methods

Image	Bicubic	Bilinear	Nearest Neighbour	[4]	[5]	[6]	[7]	Proposed Method
Boat	28.144	27.844	26.502	30.347	30.391	29.761	30.213	30.388
F 16	25.700	25.665	24.421	27.021	27.212	26.587	26.426	27.321
Fruit	25.433	25.280	24.377	26.554	26.527	25.983	26.492	26.533
Lena	30.189	29.971	28.267	33.348	33.372	32.543	33.432	33.413
Tiffany	30.083	29.726	28.455	32.012	32.128	32.092	32.217	32.112

Table 4
The Average Time to Enlargement Images (Boat, F 16, Fruit, Lena, Tiffany) from 256×256 to 512×512 Expressed in (ms) by Different Methods.

Scheme	Bicubic	Bilinear	Nearest Neighbour	[4]	[5]	[6]	[7]	Proposed Method
Average time (ms)	49	23	11	114	298	317	73	45



Figure 8. Original Lena image.



Figure 10. Reconstructed image by bilinear. PSNR = 28.842 dB.



Figure 9. Reconstructed image by bicubic. PSNR = 28.590 dB.



Figure 11. Reconstructed image by nearest neighbour. PSNR = 26.937 dB.

images between the reconstructed and original Lena generated by the four methods. Figures 13–16 are the four different images yielded by bicubic, bilinear, nearest neighbour, and the proposed method, respectively. It is obvious that Fig. 16 has the best performance because the difference is not as apparent as the others. The performance evaluation in terms of PSNR value and the difference in the image can demonstrate that the proposed method outperforms the other three methods.

To evaluate the visual performance of the proposed method, we enlarge the boat image whose size is 512×512

to 2048×2048 by the above four methods. Figure 17 is the original boat. We clip a block containing a boat hook to see the visual effect in different enlargement methods. Figure 18 is yielded by bicubic interpolation. Figure 19 shows the result by bilinear interpolation. Figure 20 shows the same part by nearest neighbour method. Figure 21 exhibits the result by our proposed method. It is obvious that the proposed method has the best visual effect among the four methods.



Figure 12. Reconstructed image by proposed method. PSNR = 31.807 dB.



Figure 13. Difference image by bicubic.



Figure 14. Difference image by bilinear.



Figure 15. Difference image by nearest neighbour.

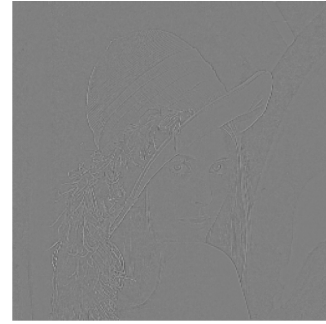


Figure 16. Difference image by the proposed method.



Figure 17. Original boat image (512×512).



Figure 18. Enlarged image by bicubic interpolation.

5. Conclusion

Some of the digital cameras provide the functionality of image enlargement, and most of them use nearest neighbour. In this paper, we achieve the goal of the image enlargement by Lagrange interpolation. The experimental results also demonstrate that the Lagrange polynomial can yield a



Figure 19. Enlarged image by bilinear interpolation.

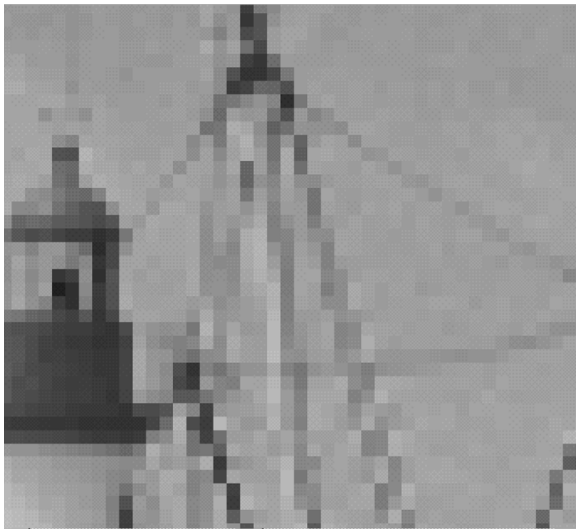


Figure 20. Enlarged image by nearest neighbour.



Figure 21. Enlarged image by the proposed method.

better quality than other methods when the visual quality and computation burden are considered. The Lagrange interpolation to enlarge the image is simple and easy to implement. Therefore, the application of Lagrange polynomial to image enlargement in this paper is successful.

Acknowledgement

The author expresses his appreciation to the editor and the anonymous reviewers for their valuable comments. This research is supported partially by the National Science Council of Taiwan under the grant number NSC 98-2221-E-309-007.

References

- [1] M. Abramowitz & I.A. Stegun, *Handbook of mathematical functions with formulas, graphs, and mathematical tables*, 9th ed., (New York: Dover, 1972).
- [2] H. Jeffreys & B. Jeffreys, *Methods of mathematical physics*, 3rd ed., (England: Cambridge University Press, 1988).
- [3] <http://mathworld.wolfram.com/LagrangeInterpolatingPolynomial.html>
- [4] K.H. Chung, Y.H. Fung, & Y.H. Chan, Image enlargement using fractal, *Proc. of IEEE International Conference on Acoustics, Speech, and Signal Processing*, Hong Kong, China, 2003, 273–276.
- [5] H. Kawano, N. Suetake, B. Cha, & T. Aso, Image enlargement by using self-decomposed codebook and Mahalanobis distance, *IEICE TRANSACTIONS on Information and Systems (Japanese Edition)*, J91-D(8), 2008, 1983–1985.
- [6] A. Watanabe & A. Taguchi, Improvement of the image enlargement method based on the Laplacian pyramid representation, *Proc. of the 47th Midwest Symposium on Circuits and Systems*, Hiroshima, Japan, 2, 2004, 585–588.
- [7] C.H. Hsieh & R.H. Hung, Image enlargement based on Grey polynomial interpolation, *Proc. of the 11th Grey System Theory and Application Conference*, Hsin Chu, Taiwan, 2006, C2–3.
- [8] Y.G. Wu & C.H. Wu, Image vector quantization codec indices recovery using Lagrange interpolation, *Image and Vision Computing*, 26, 2008, 1171–1177.

Biography



Yung-Gi Wu received his B.S. degree in information and computer engineering from Chung Yuan Christian University, Chung-Li, Taiwan, in 1992, and his M.S. and Ph.D. degrees in electrical engineering from the National Cheng Kung University, Tainan, Taiwan, in 1994 and 2000, respectively. From 1994 to 1996, he was a Reserve Officers' Training Corps (ROTC) officer in the army. He

was an assistant professor in the Department of Computer Science and Information Engineering and the Institute of Applied Information, Leader University, Tainan, Taiwan, from 2000 to 2003. He has been an associate professor since December 2003. He is now an associate professor at the Department of Computer Science and Information Engineering, Chang Jung Christian University. His research interests include image database indexing, image processing, and nonlinear optimization.